

Quantum Mirror Symmetry via Gross-Siebert Program

Based on Joint Work with Pierrick Bousseau and Bernd Siebert

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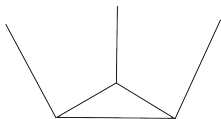
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- ① Quantum Mirror Symmetry
- ② Deformation Quantization
- ③ Proper Landau-Ginzburg Model in Gross-Siebert Program
- ④ Sketch of Proof
- ⑤ Appendices
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Classical Local Mirror Symmetry [CKYZ99][HV00][AV00][AKV02]

A-Model

Toric Calabi-Yau Threefolds
e.g. $K_{\mathbb{P}^2}$



Genus 0 Local GW of $K_{\mathbb{P}^2}$
or Logarithmic GW of $(\mathbb{P}^2, E)$

$$N_{3d}^0$$

B-Model

Landau-Ginzburg Models

$$((\mathbb{C}^*)^2_{X,P}, W)$$

$$W_{\mathbb{P}^2} = T(X + P + \frac{1}{XP})$$

Mirror Curve

$$C := \{-1 + W = 0\}$$

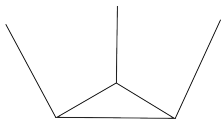
Classical Period Integrals

$$\oint \log P d \log X$$

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Already proven in great generality a long time ago.([vGRS25])

Quantum Local Mirror Symmetry (Aganagic-Cheng-Dijkgraaf-Krefl-Vafa [ACD⁺12])

Quantum A-Model

Higher Genus NS-Limit of Refined Topological String Theory of $K_{\mathbb{P}^2}$

Quantum Local Mirror Symmetry (Aganagic-Cheng-Dijkgraaf-Krefl-Vafa [ACD⁺12])

Quantum A-Model

Higher Genus NS-Limit of Refined Topological String Theory of $K_{\mathbb{P}^2}$
= Higher Genus Logarithmic GW of $(\mathbb{P}^2, E)$ with λ_g -insertion
(Bousseau [Bou20a])

$$N_{3d}^g$$

Quantum Local Mirror Symmetry (Aganagic-Cheng-Dijkgraaf-Krefl-Vafa [ACD⁺12])

Quantum B-Model

- Quantum Mirror Curve Equation

$$(-1 + \widehat{W})\psi(x) = 0, \quad \widehat{X} = e^x, \quad \widehat{P} = e^{\hbar\partial_x}$$

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$$(-1 + Te^x)\psi(x) + T\psi(x + \hbar) + Te^{\frac{1}{2}\hbar}e^{-x}\psi(x - \hbar) = 0$$

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- Quantum Periods? To use the WKB Approximation Ansatz

$$\psi(x) = \exp\left(\frac{1}{\hbar} \int^x \lambda_{\hbar}\right), \quad \lambda_{\hbar} = \sum_{i \geq 0} \lambda_i \hbar^i$$

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Theorem (Quantum Mirror Symmetry) (Bousseau-Siebert-Y.)

The higher genus NS-limit of refined topological string theory is equivalent to the quantum periods on the mirror curve for $K_{\mathbb{P}^2}$. (Our methods could be generalized to all local (toric) del Pezzo surfaces.)

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- Different choices of coordinates lead to different WKB approximations $\psi(x)$.
- However, the quantum periods, or equivalently certain monodromy of $\psi(x)$, are all the same.
- **There should be an intrinsic (coordinate-free) interpretation of the quantum periods based on the noncommutative geometry of the quantum torus and the quantum superpotential $\widehat{W}$!**

Deformation Quantization (Kashiwara et al.)

- Classical Geometry from LG Model:
 - Holomorphic Symplectic Surface X , e.g. Hori-Vafa Torus
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 - Deformation Quantization of C : a simple holonomic left DQ module supported on C with classical limit $\mathcal{O}_C$, e.g. Landau-Ginzburg DQ module

$$\mathcal{M}_{LG} := \mathcal{A}_X / \mathcal{A}_X(-1 + \widehat{W}).$$

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$$\mathcal{M}_{LG} := \mathcal{A}_X / \mathcal{A}_X(-1 + \widehat{W}).$$

- What kind of information can one extract from $(\mathcal{A}_X, \mathcal{M}_{LG})$?

(Complete) Invariants of Noncommutative Geometry Data $(\mathcal{A}_X, \mathcal{M}_{LG})$

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- Classification of DQ Algebroids $\mathcal{A}_X$: Deligne-Fedosov Class
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$$H^1(C; \mathbb{C}_\hbar^*) \longrightarrow H^2(X, C; \mathbb{C}_\hbar^*) \longrightarrow H^2(X; \mathbb{C}_\hbar^*)$$
$$\text{QC}(\mathcal{M}_C) \mapsto \mathfrak{D}(\mathcal{A}_X, \mathcal{M}_C) \mapsto \mathfrak{D}(\mathcal{A}_X) = 1$$

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$$H^1(C; \mathbb{C}_\hbar^*) \longrightarrow H^2(X, C; \mathbb{C}_\hbar^*) \longrightarrow H^2(X; \mathbb{C}_\hbar^*)$$
$$\text{QC}(\mathcal{M}_C) \mapsto \mathfrak{D}(\mathcal{A}_X, \mathcal{M}_C) \mapsto \mathfrak{D}(\mathcal{A}_X) = 1$$

- **Theorem (Bousseau-Siebert-Y.)** $\text{QC}(\mathcal{M}_{LG})$ is the same as the quantum correction of periods from WKB approximation when $\mathcal{A}_X$ is Hori-Vafa quantum torus and $\mathcal{M}_{LG}$ is the Landau-Ginzburg DQ Module supported on the mirror curve C

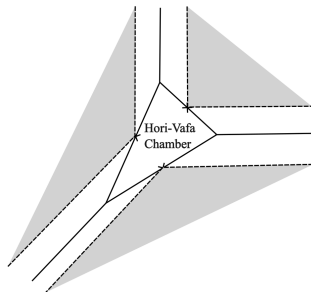
$$\mathcal{M}_{LG} := \mathcal{A}_X / \mathcal{A}_X(-1 + \widehat{W}).$$

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Proper Landau-Ginzburg Model in Gross-Siebert Program [CPS24] and Its Deformation Quantization by Bousseau [Bou20b][Bou20a]

Completion of Hori-Vafa Mirror LG Model:

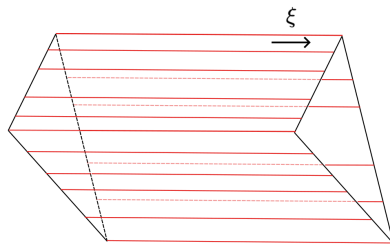
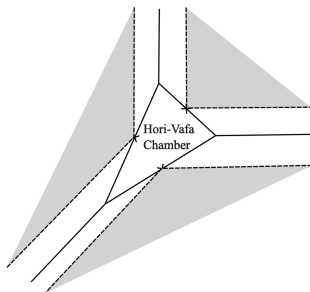
Mirror Proper LG Model (X, W) of Toric Degeneration of $(\mathbb{P}^2, E)$ together with Its Deformation Quantization $(\mathcal{A}_X, \widehat{W})$.



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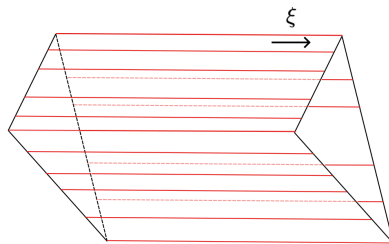
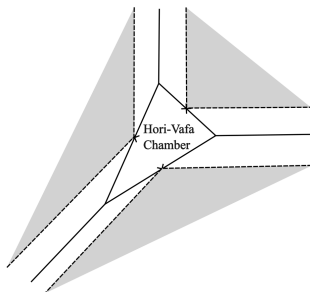


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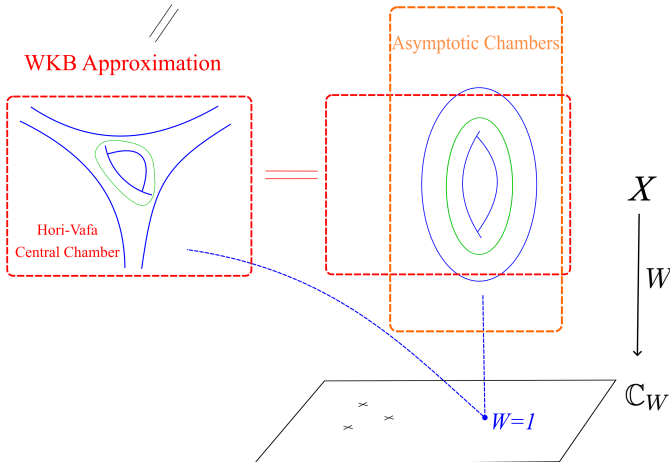
Asymptotic Chambers

Asymptotic walls are exactly log GW of $(\mathbb{P}^2, E)$ with λ_g class !

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Sketch of Proof: Compute Intrinsic Invariants in Different Coordinates

Intrinsically Defined Quantum Periods $\text{QC}(\mathcal{M}_{LG}) =$ Asymptotic Quantum Walls

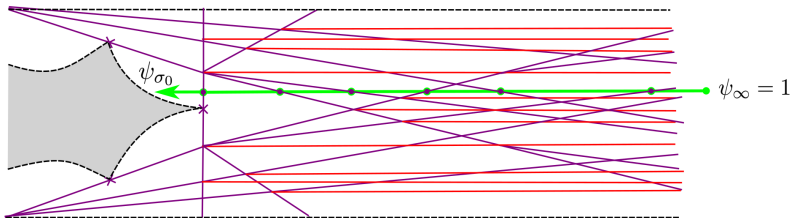


Thank You So Much!

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Quantum Open Mirror Symmetry

Asymptotic wavefunction is $\psi_\infty = 1$ after imposing the quantum open mirror map in [GRZZ25].



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